

public-key encryption 95
 public/private chat 203
 publishing, on social networks 203
 punch cards 81, 98, 99, 122
 Python 118, 130–1

Q

Quality Assurance (QA) testers 243
 Quantum Computing Language (QCL) 141
 qubits 141

R

R 140
 railway maps 72
 RAM (random-access memory) 34, 43, 49, 63, 81
 ransomware 157
 Raspberry Pi 54, 55, 123
 rasterization 90
 Real-time Streaming Protocol (RTSP) 148
 Real-time Transport Protocol (RTP) 148
 recycling 49, 164, 217
 regression tests 173
 remixing 69
 removable hard drives 37
 REPL (Read-Evaluate-Print Loop) 133
 representational state transfer (REST) 125
 research and development 242
 resolution 91
 resource sharing 145, 153
 reviews, online 196
 “Right to be Forgotten” law 199
 ring topology 147
 Ritchie, Dennis 110, 126
 robots 41, 224
 home 59
 toys 61
 ROM (read-only memory) 43
 rootkits 156
 routers 37, 151
 routing 144, 149
 RTOS (real-time operating systems) 45
 Ruby 118, 132–3
 runtime errors 115
 Rust 140

S

satellites 29, 144, 151
 saving work 15
 scaling 170
 scams 159
 scanners 48
 Schickard, William 31
 Schrems, Max 179
 science 238
 Scratch 119, 136–7
 ScratchJr 136
 screen breaks 180
 scripting languages 133
 scripts 134, 137, 166
 SD memory cards 15
 search engines 20–1, 151, 211
 setting a default 20
 search terms 21, 105
 searching, effective 21
 Searle, John 237
 Secure Socket Layer (SSL) 95
 security
 APIs 125
 cloud storage 153
 cyber 22–3
 cyberbullying 188–9
 data packets 149
 hacking and privacy 190–1
 JavaScript 167
 malware 156–7
 networks 145
 online 185, 186–7
 selectors 165
 self-driving cars 102
 self-harm 186
 selfies 195
 semantics 163
 sensors 57, 61, 64, 65, 224, 226
 servers 145
 Shannon, Claude 86
 sharing 198–9
 shopping 64
 Simple Mail Transfer Protocol (SMTP) 148
 Skype 184
 sleep patterns 180, 226
 slide rules 100
 slideshows 18
 smart devices 58–9, 129
 smartphones 19, 52–3, 58, 151, 180, 184, 200, 225, 226
 smartwatches 56, 57
 social engineering 22, 190
 social media 29, 73, 89, 180, 182, 194–5
 apps 200–1
 bubbles 206–7
 hackers 23
 platforms 196–7
 safety 187
 shared content 198–9
 social networks 19, 73, 194, 197
 gaming and 204–5
 using 202–3
 social publishing platforms 196
 software 14, 41
 application 16–17
 developing and designing 168–9, 241
 errors 114–15
 maintenance and support 174–5
 planning ahead 170–1
 rental 153
 system 16
 testing 172–3
 Software Development Life Cycle (SDLC) 168, 169
 software engineering 240
 solid-state devices (SSDs) 81
 solutions, multiple 69
 Sonic Pi 133
 sound 25, 40
 sound waves 92
 source code 111
 sources 211
 space probes 45, 218
 spaghetti code 170
 spam 156
 speakers 49
 spell-checkers 221
 spreadsheets 17
 sprints 169
 spying devices 65
 spyware 156
 SQL (Standard Query Language) 88, 140
 star topology 147
 steam 205
 STEAM 238
 steganography 90
 STEM 238
 stereoscopic vision 228
 storage
 cloud 152–3
 devices 49
 stored program computers 32
 strategy design pattern 171
 streaming 154–5
 strong ties 206, 207
 Stroustrup, Bjarne 127

stylesheets 164
 sub-problems 70, 71
 substitution ciphers 94
 supercomputers 33
 support desks 174
 switches 80, 83
 switching pages (Kodu) 139
 syntax 166
 errors 114
 system administrators 242

T

tablets 52–3, 180, 200
 Tabulating Machine 99
 tags 163
 TCP 154
 technical writers 243
 technology 238
 televisions 28
 testing, software 168, 172–3
 text, display 40
 texting 184
 thinking skills 69
 tongue, seeing with 235
 topology 147
 Tor network 158–9
 Torpedo Data Computer (TDC) 100
 touchscreen 52–3, 210
 toys, digital 60–1
 traffic control 64, 227
 transgender people 219
 transistors 38, 80, 83
 translation 110–11, 118
 Transmission Control Protocol (TCP) 149
 transposition ciphers 94
 trees 140
 trending tools 89
 trojans 156, 157
 trolling 187, 205
 truth tables 86, 87
 Turing, Alan 95, 237
 Turing test 237
 tutorials 174
 Twitter 19, 125, 207
 two-factor authentication (2FA) 191, 201
 TypeScript 140

U

UDP 154
 unfriending 202
 Unicast 154